

PEDALSHIELD

Ride private. Earn shielded.

Integrated Roadmap: Free App + Edge-Node Hardware

The model: a free phone app wins reach and demand; a BLE-tethered on-bike edge node is the premium tier that delivers battery-free always-on tracking, hardware-grade anti-cheat, and trustless verified rides. Three phases, sequenced.

Live: pedalshield.app · Apple Developer account in hand · Edge node in development
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1. The model: free app + premium node

Mass adoption must never be gated behind hardware. So the product splits cleanly:

- **Free app = reach + demand.** A genuinely good private ride tracker (route never leaves the phone) that pays small shielded ZEC. This is how Pedalshield spreads and how it generates shielded demand for Zcash. Phone-only tracking + on-device verification.
- **BLE edge node = premium tier.** A phone-tethered on-bike device for power users. It pairs over Bluetooth (phone still does GPS + upload), so it stays cheap, small, and fast to ship — while adding what software can't:
- **Battery-free always-on tracking** (the device, not the phone, carries the sensing load — solving the #1 reason trackers get deleted),
- **Hardware-grade anti-cheat** (tamper-resistant sensors beat phone-only spoofing),
- **A path to trustless rides** (hardware attestation → on-device ZK proof-of-distance), which unlocks bigger sponsors and institutions.

Why this wins: the app gets you to everyone; the node deepens your most valuable riders (higher retention, higher ARPU) and builds a moat Strava cannot copy — because Strava is a phone app and you are a privacy hardware+software platform.

2. The three phases, sequenced

Phase	Focus	Outcome
1 • Now-90 days	Ship the FREE iOS app (Apple account ready). Phone-only tracking + on-device verify + small rewards. Win the privacy/commuter niche city-by-city. Waitlist → TestFlight → App Store.	App live in the App Store; first real users; retention signal; demand proven.
2 • Months 3-9	Edge-node crowdfunding pre-sale (Crowd Supply and/or Kickstarter). The waitlist + app users are the prelaunch list. Validate demand, pre-fund the first manufacturing run non-dilutively.	Funded production run; committed buyers; community; non-dilutive capital.
3 • Months 9-18+	Manufacture + ship the node. Launch premium tier. Roll out hardware-attested → ZK trustless claims. Android app. Begin the miner / node-oracle flywheel.	Hardware shipping; premium ARPU; trustless claims; self-funding path active.

The loop that ties it together: the live waitlist (pedalshield.app) feeds the app beta; app users become the crowdfunding backers; backers become premium hardware riders; premium revenue + crowdfunding + grants fund the node/oracle and the miner flywheel.

3. Phase 1 — ship the free iOS app

You already have the Apple Developer account, so this is mechanical. Critical-path steps:

- **Production build:** move the Expo dev build to a release build via EAS Build; configure app icon, splash, bundle id, version.
- **App Store Connect:** create the app record; fill metadata, keywords (ASO), support URL, age rating.
- **Privacy nutrition labels — your weapon:** most apps list pages of data collection. Yours truthfully says “Data Not Collected.” Make that a headline screenshot, not fine print.
- **Screenshots + preview:** hero the privacy promise, the ride flow, and the real shielded payout with txid.
- **TestFlight:** invite the warmest waitlist segment; gather crash + battery telemetry before public release.
- **Submit for review:** flag that rewards are small ZEC payouts (crypto rewards), have your privacy basis ready; iterate on any rejections.

Parallel must-fix: battery-safe background tracking (auto start/stop from motion, efficient sampling).

This is the retention make-or-break for an always-on app — and the problem the edge node ultimately offloads in Phase 3. Android (Play, \$25) follows once iOS retention looks healthy.

4. The edge node — what it is, and the economics

A small BLE-tethered device that mounts on the bike (true to the IntelliGrip name). The phone handles GPS and upload; the node carries sensing, attestation, and (later) the ZK proof. Phone-tethered keeps the bill of materials and price low and gets you to market faster than a standalone cellular device.

Directional unit economics (illustrative — confirm with real BOM + manufacturer quotes):

Item	Low vol.	At scale
BOM (MCU/BLE, IMU, secure element, battery, enclosure)	~\$25–40	~\$15–22
Landed cost (BOM + assembly + cert amortized + shipping)	~\$45–60	~\$28–35
Retail price	\$99–149	\$99–129
Gross margin per unit	~50–60%	~65–75%

Plus recurring: node owners convert to Premium (\$4.99–7.99/mo) at much higher rates (higher earn caps, trustless claims, advanced analytics). Hardware lifts both one-time revenue and subscription ARPU.

Caveat: these are planning figures; certification (FCC/CE/IC), tooling, and support costs are real and must be quoted before committing.

5. Phase 2 — the crowdfunding pre-sale

Hardware's best first move is a pre-sale: it validates demand, pre-funds the first run **non-dilutively**, and builds community. Two fits:

- **Crowd Supply** — purpose-built for open/privacy hardware; the audience overlaps your thesis exactly; ongoing storefront after the campaign. Lead here.
- **Kickstarter** — broader reach and social proof; consider as a parallel or follow-on for mainstream visibility.

Campaign shape:

Element	Plan
Prelaunch list	Your pedalshield.app waitlist + app users. Build a “notify me” reservation page before going live — the size of that list predicts the campaign.
Funding goal	Set to cover the first manufacturing run + certification + fulfillment — not vanity. A goal you clear in days signals momentum.
Reward tiers	Early-bird node; node + 1yr Premium bundle; “founding rider” (node + perks + name in credits); 2-pack for clubs.
Story	The privacy moat + the working mainnet app (real txids) + the honest roadmap. Show the app already exists — de-risks the hardware ask.
Delivery honesty	Publish a realistic ship date with buffer; communicate monthly. Late/over-promised hardware is the classic crowdfunding failure — under-promise.

6. How it funds the Zcash mission

This roadmap is also the funding engine for the node/oracle and the miner flywheel from the investor plan:

- **Crowdfunding** (non-dilutive) pre-funds hardware and proves demand.
- **App revenue** (Premium + gear affiliate + corporate wellness + sponsor pools) builds toward covering reward burn.
- **ZCG grant** funds the open-source public goods (spend pipeline, FROST ceremony, ZK proof-of-distance).
- **Zodler pre-seed** (SAFE, no token, no yield) covers team + treasury seeding + the infra bridge.

Stacked, these carry you to the point where **hardware margin + subscriptions + sponsors throw off surplus** — which funds the full-time node/oracle, then ZEC miners, facility, and electricity. The edge node makes the unit economics strong enough to actually get there.

7. Honest risks

- **Hardware execution:** BOM/cert/manufacturing/support and delivery delays. Mitigate with a proven contract manufacturer, conservative ship dates, and crowdfunding only to a fundable goal.

- **App retention / battery:** background tracking must be battery-safe; the node offloads this later but Phase 1 must nail it on the phone.
- **Crowdfunding risk:** over-promising. Show the working app to de-risk; communicate relentlessly.
- **Platform / ZEC volatility / fraud / regulatory:** as covered in the investor plan — caps, repricing, layered anti-cheat, counsel.

8. Milestones

- **90 days:** free iOS app live; TestFlight retention signal; node reservation page up; crowdfunding prelaunch list growing.
- **9 months:** crowdfunding funded; first manufacturing run committed; premium tier defined; ZCG submitted; Android in beta.
- **18 months:** node shipping; premium ARPU live; hardware-attested/ZK trustless claims; full-time node/oracle; first miner partners.

Ride private. Earn shielded. Own the hardware moat.